TEAM CAPZORYX · 2026 SHENZHEN INTERNATIONAL FINTECH COMPETITION · TOPIC E

Capzoryx Copilot Execution Evidence

Proof, from real command outputs produced today, that the product runs and that the claims made in the pitch deck and technical documentation are true. Plain figures, including the ones that are not perfect.

Release: 0.7.6 (git commit `38685981d0ebdbe29beff3cdf721d3087581cd04`)
Evidence produced: 2026-10-20, between 20:39:57 and 20:44:58 (UTC+06:00, Dhaka)
Live site: https://capzoryx-copilot.vercel.app **Repository:** github.com/Pranto-404/capzoryx-copilot
Companion folder: `Capzoryx_Execution_Evidence_Logs\` holds the raw text logs and screenshots quoted here.
Rules followed: numbers are copied from command output (this PDF is generated from the logs by `submission/evidence_build/build_evidence_pdf.py`); no secrets or personal e-mail addresses in any log; the live site received read-only requests only.

Summary of checks

All commands were run from the repository root at commit 3868598 unless the command says otherwise. Green = as expected; amber = not perfect, read the section.

#	Check	Command	Result	Raw log
1	**Backend test suite**	`python -m pytest api/tests -q`	488 passed, 1 failed, 1 skipped (490 collected). The 1 failure and 1 skip are Windows-only POSIX tests.	`01_backend_pytest.log`
2	**Black-box security probe**	`python scripts/security_probe.py http://127.0.0.1:8000`	60 of 60 checks passed (local backend, throwaway database).	`02_security_probe.log`
3	**Secret scan over git history**	`python scripts/security_scan.py`	passed: 86 commits, 1146 unique file versions scanned, no credential patterns found.	`03_secret_scan.log`
4	**Dependency audits**	`npm audit --omit=dev; pip-audit -r api/requirements.lock.txt`	npm: 0 vulnerabilities. pip-audit: no known vulnerabilities.	`04a_npm_audit.log, 04b_pip_audit.log`
5	**Production build and UI strings**	`npm run build; node scripts/extract_ui_strings.cjs --check`	Build succeeded (type check + bundle); 12 JavaScript files, largest 386.2 kB. 727 interface strings up to date.	`05a_web_build.log, 05b_extract_ui_strings_check.log`
6	**On-device logic checks (6 scripts)**	`node scripts/check_*.cjs`	All 6 exit 0: assistant 36/36 supported prompts correct, 22/22 off-topic refused; site guide, record parser, currency (2), languages all passed.	`06_node_check_*.log`
6b	**Assistant held-out evaluation**	`node scripts/evaluate_local_assistant.cjs <set>`	tuned set 1 97.3% | tuned set 2 97.8% | fresh set 73.7% (fresh set is the honest figure)	`06_node_evaluate_*.log`
7	**Financing stress check**	`python submission/evidence_build/financing_stress.py 119`	14,280 randomized scenarios (119 seeds x 120), 0 failed assertions.	`07_financing_stress.log`
8	**Accessibility scan (axe-core)**	`node scripts/axe_scan.cjs`	20 of 20 views (10 pages, desktop and phone) with 0 WCAG 2.1 A/AA violations.	`08_axe_accessibility_scan.log`
9	**GitHub Actions on main**	`GET api.github.com/.../actions/runs?per_page=5`	Latest CI run on HEAD: success. 5 newest runs listed, all conclusion success.	`09_github_actions_latest_runs.log`
10	**Live site, read-only**	`GET /api/health, /api/ready, / (headers), 2 private routes`	health 200 version 0.7.6; ready 200; CSP, HSTS, frame, referrer, permissions headers present; both private routes answer 401.	`10_live_site_readonly.log`
11	**Visual walkthrough**	`python submission/deck_build/capture_screens.py`	18 screenshots of the current build on a local app (6 shown in this PDF).	`11_screenshot_walkthrough.log, screenshots\`
12	**Repository facts**	`git rev-list --count HEAD; git tag; git ls-files`	80 commits, release 0.7.6, tags submission-2026-10-20 and v0.7.5 present.	`12_repository_facts.log`

1 Backend test suite

Run 2026-10-20 20:39:57 +0600 · commit 3868598 · command `python -m pytest api/tests -q -p no:cacheprovider` · exit code 1 · raw log `01_backend_pytest.log`

488 passed, 1 failed, 1 skipped in 0:04:12 (490 tests collected). The one failure, `api/tests/test_deployment_review.py::test_private_seed_and_expired_session_recovery_preserve_data`, checks that a private credentials file gets Unix permission bits 0600; Windows does not have those bits, so it reports 0666. It is a Windows-only test failure, not a product defect, and the same suite runs on Linux in the GitHub Actions CI (section 9), where that permission check is meaningful. The one skip is the Unix process-group cleanup test, skipped on Windows on purpose. We do **not** hide either: the run was not clean on this machine.

```
...
SKIPPED [1] api\tests\test_harness_cleanup.py:10: POSIX process-group cleanup;Windows requires separate qualification
FAILED api/tests/test_deployment_review.py::test_private_seed_and_expired_session_recovery_preserve_data
1 failed, 488 passed, 1 skipped, 2 warnings in 252.63s (0:04:12)
.................................................................... [ 14%]
.....................................F.......................... [ 29%]
```

> **How to reproduce:** in a clean clone with Python 3.12: `pip install -r api/requirements.txt -r api/requirements-dev.txt` then `python -m pytest api/tests -q` (about 4 minutes).

2 Security probe: 60 hostile-input checks

Run 2026-10-20 20:40:12 +0600 · commit 3868598 · command `python scripts/security_probe.py http://127.0.0.1:8000` · exit code 0 · raw log `02_security_probe.log`

60 of 60 checks passed, 0 failed. The probe attacks a running local backend (a throwaway database, never the live site): anonymous access to private routes, requests from a foreign origin, SQL-injection style text, oversized bodies, weak passwords, login lockout, cookie flags, a second account trying to reach the first account's data, and two-step sign-in bypass attempts. Excerpt (first 8 and last 6 lines):

```
PASS anonymous /api/workspace is refused
PASS anonymous /api/account/profile is refused
PASS anonymous /api/account/workspaces is refused
PASS anonymous /api/account/members is refused
PASS anonymous /api/proposals is refused
PASS anonymous /api/audit is refused
PASS anonymous /api/fx/rate?from=BDT&to=USD is refused
PASS anonymous POST /api/assist/ask is refused
...
PASS a locked account does not accept even the right password during the lock
PASS two-step sign-in can be switched on
PASS the recovery key alone cannot bypass two-step sign-in
PASS the password was not changed by the refused recovery

60 of 60 checks passed
```

> **How to reproduce:** start the backend with `SAFELEDGER_DB=<temp file> SAFELEDGER_SIGNING_KEY=<64 random hex> python -m uvicorn api.app.main:app --port 8000`, then `python scripts/security_probe.py http://127.0.0.1:8000`.

3 Secret scan over the whole git history

Run 2026-10-20 20:39:57 +0600 · commit 3868598 · command `python scripts/security_scan.py` · exit code 0 · raw log `03_secret_scan.log`

Passed: 86 reachable commits and 1146 unique file versions were scanned for GitHub tokens, provider API keys, private keys and AWS key ids; none found. The scanner never prints a match value. Limit (printed by the tool itself): it only knows these patterns and does not prove that no secret of any other kind exists.


```
{"status": "passed", "reachable_commits": 86, "unique_history_blobs_scanned": 1146, "known_patterns": ["github_token", "prov
ider_key", "private_key",
```


> **How to reproduce:** from the repo root: `python scripts/security_scan.py` (about 2 minutes).

4 Dependency vulnerability audits

Run 2026-10-20 20:40:00 +0600 · commit 3868598 · command `npm audit --omit=dev` · exit code 0 · raw log `04a_npm_audit.log`

No known vulnerabilities in either list. `npm audit --omit=dev` (the web app's shipped packages) says "found 0 vulnerabilities". `pip-audit -r api/requirements.lock.txt` (the backend's pinned packages, checked against the public vulnerability database) says "No known vulnerabilities found". The second command ran with log `04b_pip_audit.log` at 2026-10-20 20:40:59 +0600.

```
$ npm audit --omit=dev
found 0 vulnerabilities

$ python -m pip_audit --progress-spinner off -r api/requirements.lock.txt
No known vulnerabilities found
```

> **How to reproduce:** `cd web && npm ci && npm audit --omit=dev ; pip install pip-audit && pip-audit -r api/requir ements.lock.txt` . Results change as new advisories are published.

5 Production build and interface strings

Run 2026-10-20 20:40:59 +0600 · commit 3868598 · command `npm run build` · exit code 0 · raw log `05a_web_build.log`

Build succeeded (TypeScript check, then Vite production build, 358 modules). The largest scripts are loaded only when needed; the main application file is 267.9 kB (84.9 kB compressed). The check printed "ui strings up to date: 727", meaning every translatable interface string is in the committed source list.

File	Size (kB)	Compressed (kB)
`assets/jspdf.es.min-iQe_t0jB.js`	386.25	126.43
`assets/index-VZvLReXi.js`	267.94	84.90
`assets/html2canvas.esm-DXEQVQnt.js`	201.04	47.43
`assets/index.es-BVrdq61J.js`	159.01	53.08
`assets/Assistant-zRJ-yplv.js`	138.64	48.07
`assets/Workspace-Ccmy-3MC.js`	74.72	24.00

Six largest of 12 JavaScript files, copied from the build log; the stylesheet is 81.71 kB.

```
$ node scripts/extract_ui_strings.cjs --check
ui strings up to date: 727
```

> **How to reproduce:** `cd web && npm ci && npm run build` ; then `node scripts/extract_ui_strings.cjs --check` from the repo root.

6 On-device logic checks

Run 2026-10-20 20:40:49 +0600 · commit 3868598 · command `node scripts/check_local_assistant.cjs` · exit code 0 · raw log `06_node_check_local_assistant.log`

All six scripts exit 0. They run in Node with no network. The assistant check: 36 of 36 supported questions answered correctly, 0 wrong, 22 of 22 off-topic questions refused, 0 external calls.

Script	Last line of its real output
`scripts/check_site_guide.cjs`	site guide: 25 prompts answered on the device; off-topic still refused
`scripts/check_record_parser.cjs`	record parser: 13 sentences and the negative cases passed
`scripts/check_currency_view.cjs`	Reporting currencies: exact precision, negative rounding, large values, invalid rates, persistence validation and workspace isolation passed.
`scripts/check_currency_ui.cjs`	Currency parsing and formatting: 4 precision classes and limits passed
`scripts/check_i18n.cjs`	i18n helpers: 37 languages, merge rule, language detection and default currency passed
`scripts/check_local_assistant.cjs`	supported 36, correct 36, wrong 0, off-topic refused 22/22, security prompts 8, external calls 0

> **How to reproduce:** from the repo root after `npm ci --prefix web`: `node scripts/check_local_assistant.cjs` (and the five other `scripts/check_*.cjs` files).

6b Assistant on questions it had not seen (held-out sets)

Run 2026-10-20 20:40:49 +0600 · commit 3868598 · command `node scripts/evaluate_local_assistant.cjs heldout_eval_v3` · exit code 0 · raw log `06_node_evaluate_heldout_eval_v3.log`

The honest number is 73.7% (70 of 95) on the fresh set. The two earlier sets score 97.3% and 97.8%, but the assistant was tuned against them, so they overstate real performance. The fresh set was written after tuning; it shows 6 confidently wrong answers and weaker results on security-and-authority questions. Off-topic prompts were refused in every set but one prompt in set 2 that was not recognised as off-topic. Failures are listed in the raw logs, not removed. The test sets are small and hand-written, not a statistical benchmark.

Set	Overall	Answerable	Unanswerable	Off-topic	Security	Confidently wrong	Network calls
tuned set 1 (heldout_eval)	**107/110 = 97.3%**	59/60	6/8	25/25	17/17	0	0
tuned set 2 (heldout_eval_v2)	**91/93 = 97.8%**	47/47	7/8	24/25	13/13	0	0
fresh set (heldout_eval_v3)	**70/95 = 73.7%**	36/50	4/8	25/25	5/12	6	0

> **How to reproduce:** `node scripts/evaluate_local_assistant.cjs heldout_eval, ... heldout_eval_v2, ... heldout_eval_v3` (no `--write`, so nothing in the repo changes).

7 Financing stress check (randomized scenarios)

Run 2026-10-20 20:43:06 +0600 · commit 3868598 · command `python submission/evidence_build/financing_stress.py 119` · exit code 0 · raw log `07_financing_stress.log`

14,280 randomized scenarios, 0 failed assertions. Each round builds a random business (random balance and scheduled payments) and 1 to 10 random lender quotes, then requires: no crash, repayments that add up exactly to principal plus interest, fitting offers ranked first, costs in order, and the same answer when the quotes are shuffled. Random terms outside the allowed range must be refused, and are. The committed test uses one seed (120 rounds); this run repeats it with 119 different seeds. The test file is `api/tests/test_financing_fuzz.py`; it also ran inside check 1.

```
seeds run                   : 119
rounds per seed             : 120
randomized scenarios (rounds): 14280
seeds with a failed assertion: 0
seconds                     : 63.2
```

> **How to reproduce:** `python submission/evidence_build/financing_stress.py 119` (about 1 minute; no network or database).

8 Accessibility scan (WCAG 2.1 A and AA)

Run 2026-10-20 20:44:53 +0600 · commit 3868598 · command `node scripts/axe_scan.cjs http://127.0.0.1:5173` · exit code 0 · raw log `08_axe_accessibility_scan.log`

20 of 20 views clean. axe-core scanned the sign-in page and nine workspace pages at desktop width (1366 px) and phone width (390 px) on a local app using Chrome. Automated scanning finds many but not all accessibility problems; it is not a substitute for testing with assistive technology.

```
ok   desktop sign-in
ok   desktop Overview
ok   desktop Add records
ok   desktop Invoices
ok   desktop Cash flow
ok   desktop Loans
...
ok   phone Assistant
ok   phone Approvals
ok   phone Activity
ok   phone Settings
```

> **How to reproduce:** start the backend and `npm run dev --prefix web`, then `SAFELEDGER_BROWSER_CHANNEL=chrome node scripts/axe_scan.cjs http://127.0.0.1:5173`.

9 GitHub Actions results for main

Run 2026-10-20 20:44:33 +0600 · commit 3868598 · command `GET https://api.github.com/repos/Pranto-404/capzoryx-copilot/actions/runs?per_page=5` · exit code n/a · raw log `09_github_actions_latest_runs.log`

The newest CI run on the submitted commit finished with conclusion success. Times are UTC as returned by GitHub. This PDF's commit (3868598) is the head of main for runs 1 and 2.

Workflow	Commit	Status	Conclusion	Created (UTC)
Keepalive	`3868598`	completed	success	2026-10-20T12:53:10Z
CI	`3868598`	completed	success	2026-10-20T12:32:52Z
CI	`ca7b4af`	completed	success	2026-10-20T12:25:42Z
CI	`63045ad`	completed	success	2026-10-20T12:19:18Z
CI	`14bd252`	completed	success	2026-10-20T12:12:05Z

> **How to reproduce:** `curl https://api.github.com/repos/Pranto-404/capzoryx-copilot/actions/runs?per_page=5` (public, no login) or open the repository's Actions tab.

10 Live site: read-only checks

Run 2026-10-20 20:44:36 +0600 · commit 3868598 · command `GET https://capzoryx-copilot.vercel.app/api/health, /api/ready, / (headers), /api/workspace, /api/account/profile` · exit code n/a · raw log `10_live_site_readonly.log`

The deployed site answers as designed. Only five plain GET requests were made (no sign-in, no account, no load). Health and readiness return 200; the page is served with a strict content policy, HSTS, no framing, no referrer, and camera/microphone/location blocked; private routes refuse an anonymous caller with 401. The deployed build id equals the head commit of main.

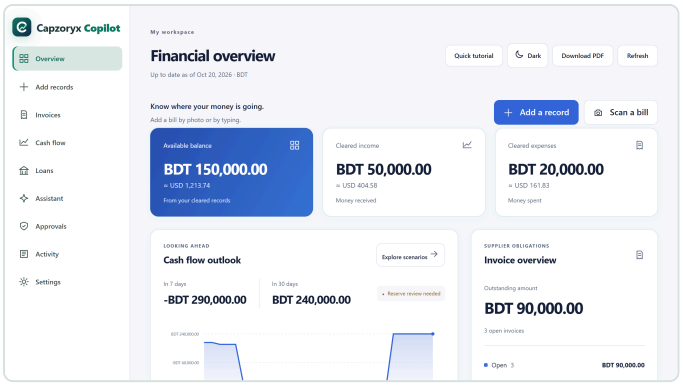
```
GET /api/health -> HTTP 200
{"status":"ok","version":"0.7.6","build_commit":"38685981d0ebdbe29beff3cdf721d3087581cd04","simulation_only":true,"releas
e":"accounts-and-assistant","
GET /api/ready -> HTTP 200
{"status":"ready","database":"libsql","schema_version":1}
GET / -> HTTP 200  (body 1247 bytes, not stored)
Security-relevant response headers:
  content-security-policy: default-src 'self'; script-src 'self'; style-src 'self'; img-src 'self' data:; connect-src 'sel
f'; frame-ancestors 'none';
  strict-transport-security: max-age=31536000; includeSubDomains
  x-frame-options: DENY
  referrer-policy: no-referrer
  permissions-policy: camera=(), microphone=(), geolocation=(), payment=(), usb=()
  x-content-type-options: nosniff
  cross-origin-opener-policy: same-origin
GET /api/workspace (no session) -> HTTP 401  {"error":{"code":"AUTHENTICATION_REQUIRED"}}
GET /api/account/profile (no session) -> HTTP 401  {"error":{"code":"AUTHENTICATION_REQUIRED"}}
```

> **How to reproduce:** `curl -i https://capzoryx-copilot.vercel.app/api/health`, `/api/ready`, `curl -I https://capzoryx-copilot.vercel.app/`, and `curl -i .../api/workspace` (expect 401).

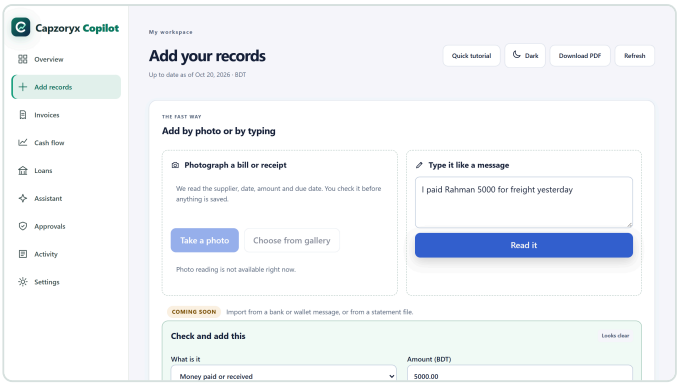
11 Visual walkthrough of the running app

Run 2026-10-20 20:44:58 +0600 · commit 3868598 · command `python submission/deck_build/capture_screens.py http://127.0.0.1:5173 D:/China Comp/Capzoryx_Submission_Final/Capzoryx_Execution_Evidence_Logs/screenshots` · exit code 0 · raw log `11_screenshot_walkthrough.log`

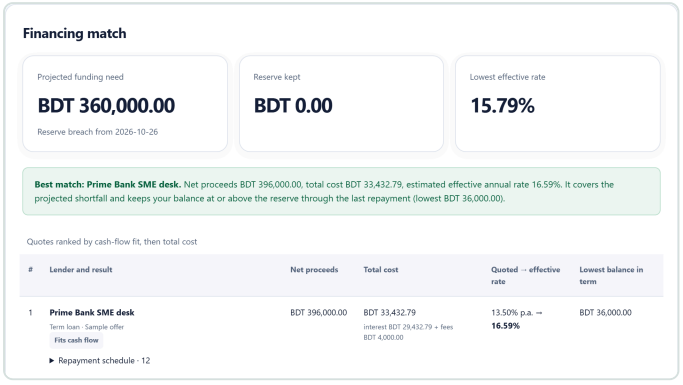
Taken today from a **local copy of the current build** (local backend on a throwaway database), signed up as a throwaway user with the built-in "Try a sample business" data; not reused from older runs. The business and numbers are fictional. Views are cropped to the top of each page; the full-size files (18 PNGs, including the phone layout `phone_overview.png`, activity, approvals, assistant, cash flow, landing pages and sharing settings) are in the logs folder `screenshots\`.



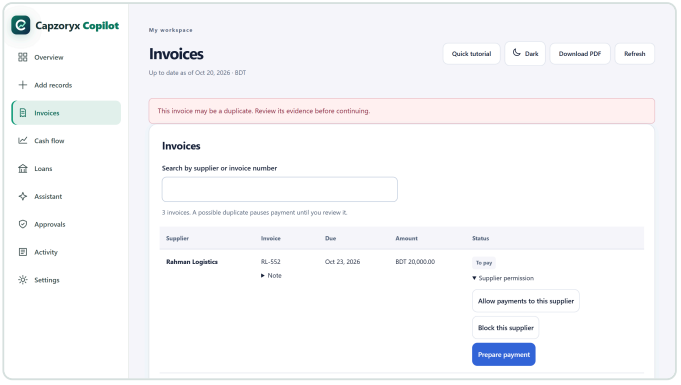
1. Financial overview of the sample business: balance, cash outlook and a projected shortfall. Real screenshot of this build.



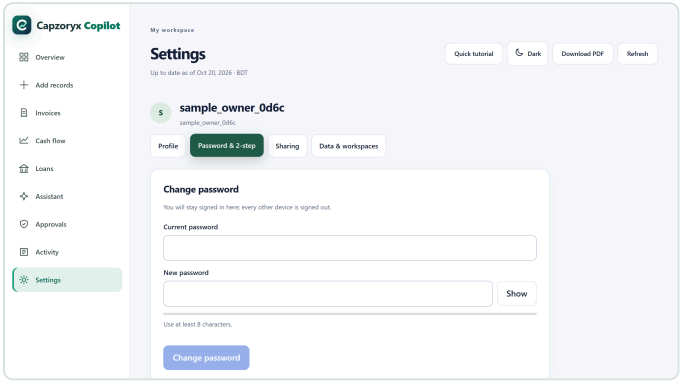
2. Add records: the sentence "I paid Rahman 5000 for freight yesterday" becomes an editable draft that is not saved until confirmed.



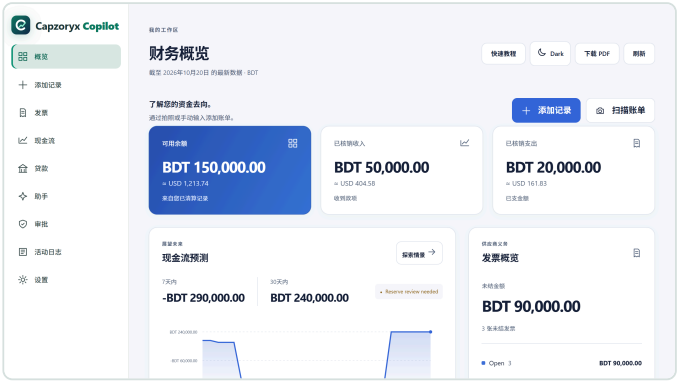
3. Loans: three lender quotes ranked against the business's own cash flow (top of the page shown).



4. Invoices: a possible duplicate invoice is flagged and payment is paused until it is reviewed.



5. Settings: password change and two-step sign-in (authenticator app).



6. The same overview in Chinese; 37 interface languages ship in total.

> **How to reproduce:** start the local backend and `npm run dev -- --port 5173 --strictPort` in web, then `python submission/deck_build/capture_screens.py http://127.0.0.1:5173 <output folder>`.

12 Repository facts

Run 2026-10-20 20:44:39 +0600 · commit 3868598 · command `git rev-list --count HEAD; git tag; git ls-files; count lines; read api/app/main.py` · exit code n/a · raw log `12_repository_facts.log`

Source of record is the git repository. Line counts are non-blank lines of tracked source (translation files, lock files, fonts and screenshots excluded); the front-end files are dense, so line counts understate the size of the web app.

```
Commit count on the current branch : 80
Branch : main
HEAD : 38685981d0ebdbe29beff3cdf721d3087581cd04
First commit date : 2026-10-17
Latest commit date : 2026-10-20
Tags (2): submission-2026-10-20 -> 3dc503b (2026-10-20); v0.7.5 -> dfc8a8f (2026-10-20)
Tag submission-2026-10-20 points at : 208539657b661bf69fc90cfee6b6f165d051303b
Tag v0.7.5 points at : 4fb35fa0fd98b1cbc42ff3119c4fc5b68d7554e5
Release in api/app/main.py : title "Capzoryx Copilot", version "0.7.6"

Tracked files: 641
Files by top-level folder:
  evidence           324
  web                79
  submission         72
  api                65
  scripts            42
  docs               29
  data               15
  (repo root)        12
  .github            2
  deployment         1

Non-blank lines of code by language (tracked files; translation files, lock files, fonts, screenshots excluded):
  JSON                          22,194 lines in 121 files
  Python                         9,078 lines in 93 files
  Markdown                       2,055 lines in 42 files
  TypeScript (React)             1,455 lines in 23 files
  JavaScript (node scripts)        663 lines in 16 files
  TypeScript                       553 lines in 12 files
  CSS                              461 lines in 1 files
  YAML                             130 lines in 3 files
  HTML                               1 lines in 1 files
  JavaScript                         1 lines in 1 files

  TOTAL                         36,591 lines

Backend tests collected: 490 tests collected in 1.78s
```

> **How to reproduce:** `git rev-list --count HEAD`, `git tag`, `git ls-files | wc -l`; counting script: `submission/evidence_build/live_and_repo.py`.

What this evidence does not show

- The test suite was not fully green on the Windows machine used here: 1 failed and 1 skipped, both Unix-specific (see section 1).
- The on-device assistant is weaker on questions it was not tuned for (section 6b, 73.7%).
- No independent penetration test was performed; the probe and scan are our own tools. No load test was run on the live site, on purpose.
- The product is a simulation-only decision-support tool: it does not hold or move money. AI features that call an online model depend on a free-tier key and were not exercised in these runs (the local backend had no model configured).
- Running the test suite and the secret scan rewrites two tracked files in the repository (a fixture manifest and a capability report); these were restored and are not part of this commit.